

I. Concept Questions (50 points, 2 points for each answer) (概念题, 50 分, 每个答案 2 分)

- Convert decimal 835 to 8421 BCD (把十进制数 835 转为 8421BCD 码):_____;
- The 2's complement of 11101000 (11101000 的补码) is:_____;
- Convert gray code 11101 to binary number (把格雷码 11101 转为二进制数) _____;
- The odd parity bit for 1100 0111 is (数据 1100 0111 的奇校验位是):_____;
- A 2-input gate produces the output shown (一个二输入门电路可得到下图输出). (A and B represent the inputs, X represents the output) (A 和 B 为输入, X 为输出) This is a(n) (这是什么门电路):_____;



(Question. 5)

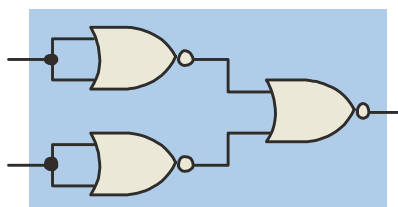
- The detailed expression for $X = A \square B$ ($X = A \square B$ 的具体表达式) is:_____;
- Convert $X = (\bar{A}+B)(A+B+C)$ to standard POS form (把 $X = (\bar{A}+B)(A+B+C)$ 转化为标准 POS):_____;
- Determine the logic expression from the truth table given below, the corresponding POS expression is (根据以下真值表得出对应 POS 逻辑表达式):_____;

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

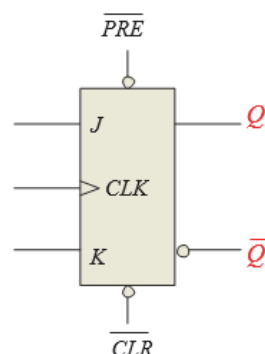
(Question. 8)

	$\bar{C}$	C
$\bar{A}\bar{B}$	1	1
$\bar{A}B$		
AB		
$A\bar{B}$	1	1

(Question. 9)

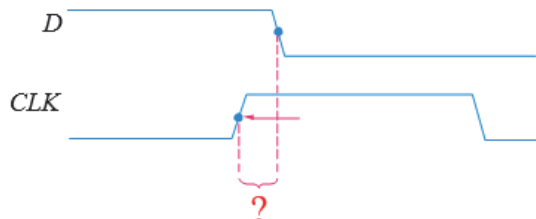


(Question. 10)

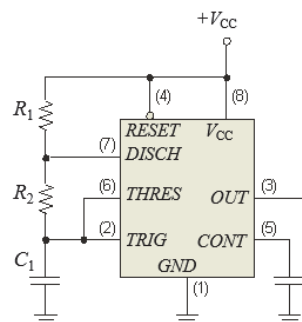


(Question. 11)

- From the Karnaugh map given, derive the simplest SOP expression (写出所示卡诺图对应的最简 SOP):_____;
- The circuit shown is equivalent to (图所示电路等效为哪种逻辑门):_____;
- For the J-K flip-flop shown, the number of inputs that are asynchronous is (在图中所示的 J-K 触发器中, 异步输入的个数为):_____;
- The time interval illustrated is called (图中的时间间隔称为):_____;
- The given circuit based on 555 timer is a (图中所示基于 555 定时器的电路为):_____;

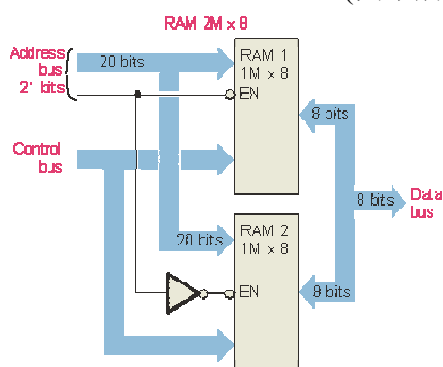


(Question. 12)

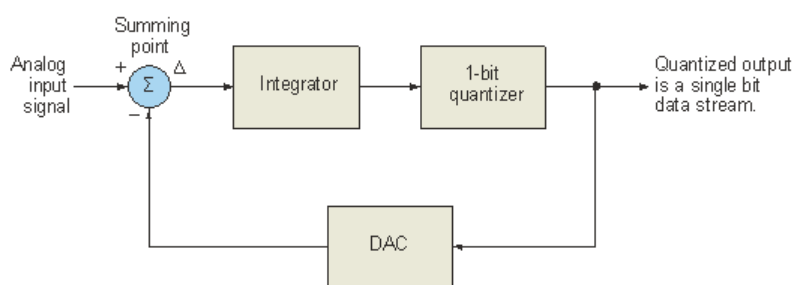


(Question. 13)

14. Assume the clock for a 4-bit binary counter is 80 kHz (假设一个 4 位二进制计数器的时钟频率为 80 kHz). The output frequency of the fourth stage (Q_3) is (第四级 Q_3 的输出信号频率为):_____;
15. The terminal count of a 4-bit binary counter is (一个 4 位二进制计数器的终端计数为):_____;
16. The delay time caused by a 4-bit serial-in/ serial-out shift register is(一个四位串入/串出移位寄存器提供的延迟时间为) :_____clock period(s)(时钟周期);
17. Among SRAM, DRAM and Flash (在SRAM, DRAM和Flash中), which cell is nonvolatile? (哪种存储单元具有非易失性?)_____; which cell is the fastest? (哪种存储单元速度最快?)_____;
18. The small triangles on the logic diagram indicate that these outputs are (逻辑框图中的小三角形表明输出是):_____;
19. To build a 256Kx4 RAM, how many 64Kx1 RAMs are needed? (需要多少片64Kx1的RAM才能构建一个256Kx4的RAM?)_____;
20. Which type of memory expansion is implemented using the circuit shown? (图中所示电路是用来实现何种内存扩展?)_____;
21. The number of comparators required in a 10-bit flash ADC is (一个 10 位快速 ADC 中所需要的比较器数目是):_____;
22. The ADC circuit shown is a (图中所示是哪种类型 ADC 电路):_____;

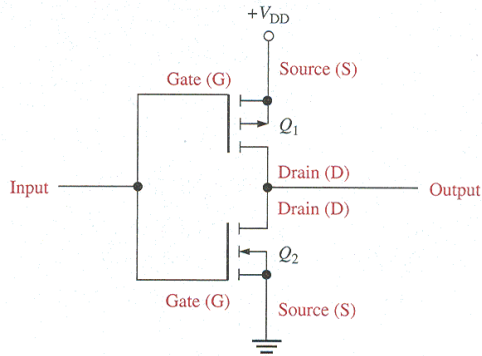


(Question. 20)

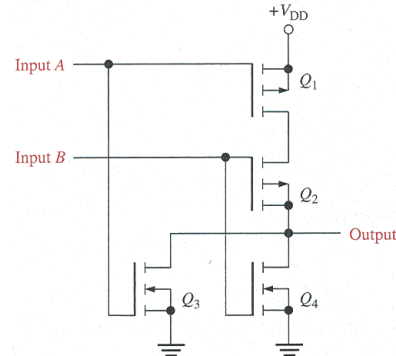


(Question. 22)

23. What is the logic function performed by the circuit given? (该电路实现的逻辑功能是?) :_____;
24. Will Q_3 be on or off for the given inputs? (给定以下输入 A、B 的组合, Q_3 是导通或截止?)
A=Low, B=High:_____; A=High, B=High:_____;



(Question. 23)



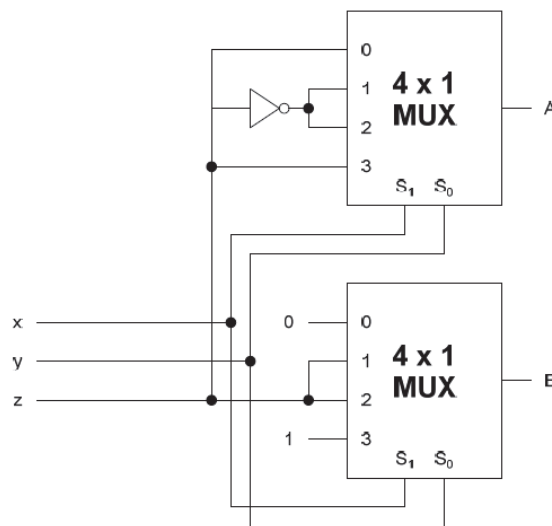
(Question. 24)

II. Design Problems (50 points) (设计题, 50 分)

Notice: For each problem, describe the details of your solution step by step.

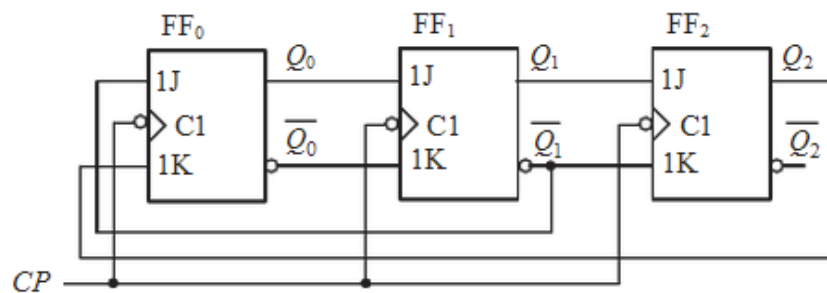
(注意: 必须详细描述各个解题步骤)

- Implement a one bit active-LOW comparator for both equality and inequality using NAND and Inverter gates (使用与非门和非门实现 1 位低电平输出有效比较器, 实现相等与不等比较). (10 points)
- Determine the outputs functions A and B of the circuit based on 4×1 multiplexer (写出所示电路输出 A 和 B 的逻辑表达式). (11 points)
 - The circuit shown in a) has the functionality of a commonly used arithmetic component (a) 中所示电路实现一个常用的算术运算单元). What is the function of the circuit and what are other names for A and B ? (电路所实现的功能是什么? A 和 B 分别代表什么输出?) (4 points)



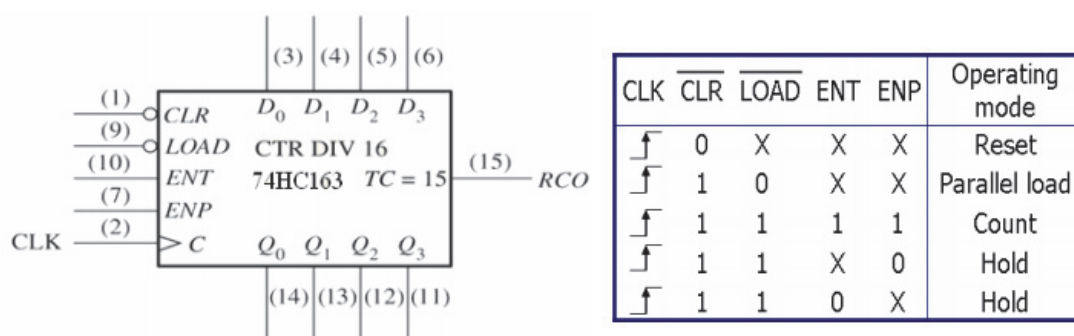
(Question. 2)

- Determine the logic function of the circuit sketched below (确定下图所示电路的逻辑功能). The detailed procedure is required. (需给出具体过程) (15 points)



(Question. 3)

4. Design a modulus-8 counter with 74HC163 (用 74HC163 设计一个模-8 计数器). The counter is required to go through the following sequence: 4, 5, 6, 7, 8, 9, 10, 11, 4, ... (要求计数器产生以下时序: 4, 5, 6, 7, 8, 9, 10, 11, 4, ...) (10 points)



(Question. 4)